

MASTER BLUEPRINT & DEVELOPMENT PLAN

GHSS Ghallanai

The complete build specification for a SaaS-grade website and installable PWA: research-backed design system, four programme microsites, admissions and results engines, role-based portals, and a sixteen-week execution roadmap.

Government Higher Secondary School Ghallanai

Mohmand District · Khyber Pakhtunkhwa · Pakistan

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1. Executive Summary

PROJECT BRIEF • GOVERNMENT HIGHER SECONDARY SCHOOL GHALLANAI

This document is the complete, build-ready master blueprint for the GHSS Ghallanai Digital Campus: a SaaS-grade website and installable PWA that will give the Government Higher Secondary School of Ghallanai a digital presence competitive with the finest college websites in Pakistan and the world. It consolidates deep competitive research, a full design system, page-by-page specifications, four functional modules, a complete technology architecture, and a sixteen-week execution roadmap into a single source of truth that a student developer can follow from the first commit to public launch.

1.1 The Institution

Government Higher Secondary School Ghallanai is a public higher-secondary institution located in Ghallanai, the administrative headquarters of Mohmand District in Khyber Pakhtunkhwa, Pakistan. The school serves the two decisive years of pre-university education: first year and second year, corresponding to grades eleven and twelve under the BISE board system. It offers four academic streams, namely ICS (Computer Science), Pre-Medical, Pre-Engineering, and Arts, which together cover the full spectrum of career ambitions held by students of the district, from medicine and engineering to computing, public service, and the humanities.

The school currently has no meaningful digital presence. In a district where mobile connectivity is widespread and families increasingly rely on smartphones and WhatsApp for information, this absence has real costs: admission inquiries go unanswered outside school hours, results and merit information circulate as unverifiable screenshots, and the achievements of students and faculty never reach the wider community. A modern website directly addresses all three problems while projecting an institutional seriousness that government schools in the merged districts are rarely credited with.

16 wks Full timeline to public launch	0 PKR/mo Launch hosting cost on free tiers	25+ Public routes in the sitemap	4 Role-based portals (SaaS layer)
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1.2 What SaaS-Grade Means for a School Website

The term SaaS-grade is used throughout this plan as a measurable quality bar, not as marketing language. It means the platform behaves like a software product rather than a static brochure: users authenticate into role-specific workspaces; data lives in a properly modelled relational database protected by row-level security; content is managed through an admin dashboard instead of code edits; the interface ships with a disciplined design system, dual bright and dark themes, and a component library; the site installs to a phone home screen as a PWA and works offline; and every page is engineered to meet SEO, accessibility, and performance budgets. Each of these qualities is specified, budgeted, and scheduled in the chapters that follow.

This standard was not invented in the abstract. It was derived from a systematic teardown of seven reference institutions across three tiers: the two benchmarks named by the school (Aitchison College and TMUC), the world's most decorated school websites (Phillips Exeter Academy and Eton College), and the institutions whose context most closely matches GHSS Ghallanai (Punjab Colleges, Islamia College Peshawar, and Edwardes College Peshawar). Chapter 2 documents what each of them does well and which of their patterns this plan adopts.

1.3 The Platform in One Paragraph

The platform is a single Next.js 16 application deployed on Vercel, with Supabase providing the PostgreSQL database, authentication, row-level security, and file storage, and Cloudinary serving as the media pipeline for photos and videos. The public website is server-rendered for SEO and styled with a custom Green and Gold design system built on Tailwind CSS 4 tokens. Four authenticated portals (student, teacher, admin, alumni) constitute the SaaS layer. The whole system is version-controlled on GitHub, deployed continuously through Vercel previews to production, and costs nothing per month at launch scale. Chapter 14 documents exact free-tier limits and the paid upgrade path.

1.4 Success Metrics

Success must be observable in numbers, not opinions. The following key performance indicators define the targets that the roadmap in Chapter 12 is engineered to hit. Baselines will be re-measured at each milestone review, and every metric maps to a specific engineering decision documented later in this plan.

KPI	Baseline today	Target at launch	Target at 6 months
Organic sessions per month	0 (no site)	1,500	6,000
Admission applications started online	0	60% of all forms	90% of all forms
Admin time to publish a notice	n/a (paper)	under 5 minutes	under 2 minutes
Lighthouse Performance (mobile)	n/a	90 or higher	95 or higher
Lighthouse Accessibility	n/a	95 or higher	100
LCP on 4G Pakistani mobile	n/a	2.0 s or less	1.6 s or less
PWA installs	0	150	600
WhatsApp notification opt-in	0	300 families	1,000 families

Table 1: Launch and six-month success metrics for the Digital Campus platform.

1.5 How to Read This Plan

Chapters 2 and 3 establish the evidence base and the vision. Chapters 4 through 6 define everything a designer and front-end developer need: information architecture, the Green and Gold design system, and page-by-page specifications. Chapters 7 and 8 define the functional modules and the technology that powers them. Chapters 9 through 11 cover the three quality frontiers that separate an adequate site from a great one: PWA behaviour, SEO, and performance, accessibility, and security. Chapters 12 through 14 are the execution layer: roadmap, testing, and deployment economics. Chapter 15 closes with maintenance and the future roadmap. Appendices A and B provide the research sources and a master build checklist that can be printed and pinned above the developer's desk.

2. Research and Competitive Analysis

EVIDENCE BASE • SEVEN INSTITUTIONS • THREE CONTINENTS

Before a single design decision was made, the leading college and school websites of Pakistan, the United Kingdom, and the United States were examined page by page: navigation architecture, page inventory, typography, animation patterns, portal systems, admissions journeys, and technology stacks. This chapter records what each reference institution does at world standard, and converts those observations into the concrete patterns the GHSS Ghallanai platform will adopt.

2.1 The Two Named Benchmarks

Aitchison College, Lahore (est. 1886)

Aitchison.edu.pk is the most prestigious school website in Pakistan and behaves like it. The homepage opens with a full-bleed video hero rather than a static banner, immediately followed by institutional announcements written with real editorial care, such as the launch of Aitchison Connect, the school's first dedicated alumni portal, released for its 140th anniversary year. The information architecture is heritage-first: About Us expands into History, Archives, the Principal's Office, Past Principals, a formal Code of Conduct, and a statement of Core Purpose, which is precisely the structure a 139-year-old institution should present. Academic life is organised into three named schools, supported by a Teaching and Learning Centre, university counselling, and university placement pages that signal outcomes rather than promises.

Two patterns are adopted directly from Aitchison. First, the alumni portal as a first-class product with its own subdomain and registration flow, which becomes the model for the GHSS Alumni Network in Chapter 7. Second, the layered typography: Aitchison pairs IBM Plex Sans for interface text with Montserrat for headings and the script face Playball for ceremonial flourishes, proving that a school site can be both institutional and warm. Its front-end stack (Bootstrap 5, Swiper 12, OwlCarousel) is dated by modern standards, which is itself a lesson: this plan specifies Tailwind CSS 4 and native CSS motion instead of a carousel-plugin stack, so the GHSS site will look current for far longer.

TMUC, Islamabad (The Millennium Universal College)

TMUC.edu.pk runs on a heavily customised WordPress education theme, and its value to this analysis is in the experience layer. The site uses scroll-triggered AOS reveal animations, odometer-style statistics counters that count up when scrolled into view, a slick carousel for programmes, magnific-popup galleries, and even a subtle custom cursor. Its navigation is the deepest in the Pakistani sample: eight schools of study expand into dozens of named degree programmes, organised in a persistent mega-menu that never leaves the user lost. The typography pairs Hind with Lexend Deca, both highly legible humanist sans-serifs.

Three patterns are adopted from TMUC. First, the mega-menu navigation pattern, which scales from today's four programmes to any future expansion without redesign. Second, the animated statistics band: counters showing placements, scholarships, and achievements are among the highest-retention elements on education sites and cost little to build. Third, the student-services mindset: Online Fee Payment, a Career Development Centre, an incubation centre, an academic calendar, and a dedicated alumni section all live in the primary navigation. The GHSS plan mirrors this by giving Campus Life, Portals, and Alumni their own top-level presence. TMUC's WhatsApp click-to-chat integration is also adopted; in Mohmand District, WhatsApp is the default communication channel.

2.2 The World Benchmark

Phillips Exeter Academy, USA (est. 1781)

Exeter.edu won a 2025 CASE Circle of Excellence award for its website redesign, making it the single most authoritative design reference for a school website today. Its homepage is narrative rather than navigational: a single line of institutional positioning (Uniting goodness and knowledge since 1781) above a large story block, with concrete proof points rendered as large typographic statistics, including a 30 million dollar financial-aid figure and a 12:1 class ratio. The signature academic programme, the Harkness method, is given a branded storytelling page of its own, teaching this plan that a school's differentiator deserves its own named page, not a bullet in a list.

Two further patterns are adopted. First, the immersive virtual campus tour (Exeter embeds a YouVisit 360-degree walkthrough): the GHSS plan schedules a photographed campus tour that loads in a lightbox and, at Phase 4, upgrades to 360-degree imagery if equipment allows. Second, the #iamExeter student-story format: short first-person accounts with names, classes, and photographs, which will become the Voices of Ghallanai section, because no statistic communicates a school better than a student explaining what it changed for them.

Eton College, UK (est. 1440)

Etoncollege.com is the reference for institutional depth managed without chaos. Its mega-menu organises more than sixty pages under eight primary sections, each with a consistent two-level structure, proving that depth and clarity are compatible when navigation is designed as a system. Functionally, Eton separates a public marketing site from a signed-in Parent Portal, runs EtonX as a separate online-learning product, maintains an Old Etonian alumni network, publishes museums and collections content, offers facilities for hire, and even serves podcasts and term dates from the same architecture. Technically, it is PWA-aware: the site ships a web manifest, apple-touch icons, and SVG crest assets, and it instruments user behaviour with Microsoft Clarity heatmaps alongside Google Tag Manager.

Three patterns are adopted from Eton. First, the strict separation of public site from authenticated portal, which becomes the architectural split between the ISR public pages and the four signed-in portals. Second, the PWA manifest discipline: GHSS ships manifest, maskable icons, and theme colours from day one rather than retrofitting. Third, behavioural analytics: Microsoft Clarity is free and will be installed at launch so that real usage data drives the post-launch improvement cycle described in Chapter 15.

2.3 The Contextual Peers

Punjab Colleges (PGC), Lahore (est. 1985)

Pgc.edu represents the largest private college network in Pakistan, and its academic model is a perfect superset of GHSS Ghallanai's: FA Humanities, FA General Science, F.Sc Pre-Engineering, F.Sc Pre-Medical, ICS, and I.Com. This validates the programme taxonomy at the heart of the GHSS sitemap. The site's most effective elements are its achievements band (142 top board positions in 2024, rendered as counters), its dedicated alumni registration journey, its student portal login in the header, and its scholarship and fee-concession pages aimed squarely at parents making financial decisions. The lesson adopted: the Admissions section of the GHSS site must speak to parents' financial questions, with fee structure, scholarships, and concessions presented as first-class pages, and results achievements must be counted, dated, and named.

Islamia College Peshawar (est. 1913)

Icp.edu.pk matters to this plan as the leading government-side reference in KPK. It operates a CMS login for content staff, an eOffice integration, online application forms for FA, F.Sc, BS, and postgraduate admissions, downloadable prospectus and fee documents, an active press-release channel, an alumni association with registration, a Quality Enhancement Cell, and a business incubation centre. What it lacks is modern front-end engineering: the public site is a legacy PHP build with dated visuals. This is the single most important competitive observation in the entire research: the strongest government institution in the province has not yet combined government credibility with modern web experience. GHSS Ghallanai can be first in the merged districts to do exactly that, at zero software cost.

Edwardes College Peshawar (est. 1900)

Edwardes.edu.pk is the closest contextual model: a century-old college in KPK with an Online Admissions e-Portal, an online merit list, an alumni registration form, notice board, societies, gallery, and e-learning links. Culturally it does something this plan explicitly adopts: it renders Urdu text in proper Nastaliq typography rather than default system fonts, and it proudly publishes Pashto Culture Day celebrations and a video tour. The GHSS design system therefore includes an Urdu typography track with Noto Nastaliq Urdu, and the content strategy schedules coverage of Pashto cultural events alongside academic news. Edwardes also demonstrates the demand for leave-application and counseling workflows in a KPK college context, reflected in the teacher-portal feature list in Chapter 7.

2.4 Feature Comparison Matrix

The matrix below compares the seven reference institutions against the GHSS Ghallanai target specification. It shows a deliberate strategy: match the global elite on experience fundamentals, exceed the national norm on transparency features such as merit lists and result lookup, and match the contextual peers on cultural fit. No government higher-secondary school in the merged districts currently offers even half of this column.

Capability	Aitchison	TMUC	Exeter	Eton	PGC	ICP	Edwardes	GHSS target
Video hero / storytelling	Yes	Yes	Yes	Partial	Yes	No	Partial	Yes
Mega-menu navigation	Partial	Yes	No	Yes	Partial	No	No	Yes
Dual bright + dark theme	No	No	No	No	No	No	No	Yes
Online admission application	Form-based	Yes	Yes	Yes	Yes	Yes	Yes	Yes + tracking
Result / merit lookup	No	Partial	No	No	Yes	Partial	Yes	Yes + analytics
Student portal	My Aitchison	Yes	Yes	Parent portal	Yes	CMS	Partial	Yes
Teacher / staff portal	Partial	Partial	Yes	Yes	Partial	eOffice	No	Yes
Admin content dashboard	Partial	Yes	Yes	Yes	Yes	Yes	Partial	Yes
Alumni network	Aitchison Connect	Yes	Yes	OEA Online	Register	Association	Form	Full network
PWA (install + offline)	No	No	No	Manifest only	No	No	No	Full PWA
WhatsApp integration	No	Yes	No	No	Partial	No	No	Yes + alerts
Urdu / Nastaliq typography	No	No	n/a	No	No	Partial	Yes	Yes

Table 2: Feature comparison of the seven researched institutions against the GHSS Ghallanai target.

2.5 Ten Principles Distilled From the Research

1. Lead with proof, not adjectives. Exeter and PGC convert claims into counted, dated, named evidence; the GHSS homepage leads with real statistics and named toppers.
2. Depth through system, not luck. Eton's sixty-page IA stays navigable because the mega-menu is engineered; navigation is designed before pages are built.
3. Separate the public stage from the signed-in workspace. Public pages sell the institution; portals run it. Different rendering strategies, different layouts, one codebase.
4. Treat admissions as a journey, not a PDF. Every reference institution replaced paper forms with tracked online applications; GHSS adds WhatsApp status alerts on top.
5. Alumni are a product, not an afterthought. Aitchison Connect shows an alumni portal can be a headline feature; the GHSS Alumni Network launches with the same prominence.
6. Animation is seasoning, not the meal. TMUC's reveals and counters delight at low cost; every animation must respect reduced-motion and add comprehension, never decoration alone.
7. Local culture is a feature. Edwardes' Nastaliq Urdu and Pashto Culture Day coverage are adopted as first-class design and content requirements for Ghallanai.

8. Free infrastructure is real infrastructure. None of the Pakistani references needed paid hosting to look world-class; the GHSS stack runs on Vercel, Supabase, and Cloudinary free tiers.
9. Performance is a feature parents can feel. Most visitors will arrive on mid-range Android phones over 4G; every design decision is tested against a 2-second LCP budget.
10. Ship a PWA from day one. No reference institution has a true installable offline-capable app; this is the single clearest way GHSS Ghallanai can leapfrog every peer in the sample.

2.6 The Opportunity Gap

The research exposes a specific opening. The elite private institutions own prestige storytelling but rarely publish raw transparency features such as public merit lists, and none of them serves a frontier-district community. The government institutions in the sample have credibility and process maturity but visually dated platforms that students compare unfavourably with everything else on their phones. GHSS Ghallanai sits at the intersection: a government institution with community trust, building on 2026-era technology, with full transparency features and a free-tier budget. Executed to this plan's specification, the finished platform will be, to the best of the research's knowledge, the most capable website of any government higher-secondary school in the merged districts of KPK, and a reference point for the province.

3. Vision, Scope and Success Metrics

WHO WE ARE BUILDING FOR · WHAT IS IN · WHAT IS OUT

A platform succeeds when it serves named people with specific jobs to be done. This chapter fixes the five personas the Digital Campus must serve, draws the boundary between the launch scope and the future roadmap, and binds the vision to the measurable targets already summarised in Chapter 1.

3.1 Vision Statement

By the end of the academic year, every family in Mohmand District will be able to discover GHSS Ghallanai, apply for admission, follow results, and stay connected to the school from a phone in their pocket, in English and Urdu, at any hour, on any network, without standing in a single queue.

The vision deliberately emphasises the phone, the language, and the queue. Research for this plan found that the mobile share of Pakistani web traffic is overwhelmingly dominant, that Urdu renders poorly when ignored, and that every administrative process in the contextual peer group still requires physical presence. Those three constraints define the product problem this platform solves. Everything else in this document exists to make that sentence true on launch day.

3.2 The Five Personas

Persona	Profile	Primary jobs to be done	Success looks like
Prospective student	Ayesha, 15, topped her matric exam in Ghallanai, owns a shared family Android phone	Choose between ICS, Pre-Medical, Pre-Engineering, and Arts; understand eligibility; apply online; check merit list	Applies in under 15 minutes on one bus ride of connectivity; installs the PWA to track her application
Parent	Haji Saifullah, 48, shopkeeper in Mohmand, WhatsApp-first, reads Urdu comfortably	Verify the school is legitimate and serious; understand fees and documents; follow admission and result dates	Finds fee structure and dates in two taps; receives WhatsApp alert when the merit list is published
Teacher	Mr. Khan, subject specialist, teaches 1st-year Physics, comfortable with basic apps	Take attendance quickly; upload assignments; enter marks; answer student questions after hours	Marks 40 students present in under a minute; students see the assignment the same evening
Alumnus	Naveed, class of 2019, now studying at university in Peshawar	Stay connected; find batchmates; guide current students; be recognised	Registers in two minutes, appears in the directory, signs up as a mentor
Administrator	The principal and office staff, custodians of records and communications	Publish notices without a developer; review applications; generate merit lists; keep data safe	Publishes a notice in under five minutes; exports the merit list as PDF in one click

Table 3: Persona summaries with jobs to be done and their definition of success.

Each persona maps to a specific surface of the platform: Ayesha lives in the public pages and the admissions flow, Haji Saifullah lives in WhatsApp and the fee pages, Mr. Khan lives in the teacher portal, Naveed lives in the alumni network, and the administrator lives in the admin dashboard. The design system in Chapter 5 and the module specifications in Chapter 7 reference these personas by name whenever a trade-off needs a tiebreaker.

3.3 Scope Boundary

A sixteen-week, single-developer roadmap demands ruthless scope discipline. The table below is the contract: everything in the left column is engineered, tested, and launched within the plan; everything in the right column is explicitly deferred to the post-launch roadmap in Chapter 15. Scope additions during development must displace something of equal size, or be deferred; this rule is restated in the risk register in Chapter 12.

In scope at launch	Explicitly deferred (v2 and beyond)
Public website: 25+ routes, 4 programme microsites, full design system, dual themes	Full Urdu and Pashto localisation of every page (launch ships key pages and Nastaliq typography only)
Online admissions with admin review queue and public merit lists	Online fee payment gateway integration (launch publishes fee structure and bank details)
Result lookup by roll number with toppers wall and result analytics	Direct BISE API integration (launch uses supervised CSV import verified by the exam branch)
Student and teacher portals: attendance, assignments, timetable, notices, gradebook	Full LMS with video lessons, quizzes, and auto-grading (launch ships assignments, resources, and submissions)
Alumni network with verified registration, directory, stories, and mentorship	Job board and CV marketplace for alumni
Admin SaaS dashboard with content management and analytics	Multi-campus / multi-school tenancy
Complete PWA: offline shell, install, web push, WhatsApp + email alerts	Native Android app (the PWA covers installability)
SEO foundation: structured data, sitemaps, local listings, CWV budgets	Paid advertising campaigns and boosted content

Table 4: Launch scope contract: in-scope capabilities versus deferred roadmap items.

3.4 Guiding Principles

- Mobile-first is non-negotiable. Every design is drawn at 360 pixels wide first and enhanced upward, because that is the device the majority of Mohmand families own.
- Bilingual dignity. English is the primary site language, but Urdu appears with proper Nastaliq typography wherever it is shown, never as an afterthought font.
- Zero-cost operation. The platform must run entirely on free tiers at launch scale, with a documented, reversible path to paid tiers only when usage genuinely outgrows them.
- Transparency as strategy. Merit lists, fee structures, and result statistics are published openly; trust is the school's strongest asset in the community.
- Accessibility is inclusion. WCAG 2.2 AA compliance is a launch gate, not an enhancement, because education sites serve users with the widest possible range of abilities.
- The student developer must be able to maintain it alone. Every architectural choice optimises for one motivated builder with a laptop and intermittent electricity, which is why managed services replace self-hosted servers throughout.

4. Information Architecture and Sitemap

NAVIGATION MODEL · ROUTE MAP · USER JOURNEYS

The information architecture converts the research patterns of Chapter 2 into the concrete route map of the platform. It is organised as a two-tier system: a public stage of six primary sections modelled on Eton's mega-menu discipline, and an authenticated portal tier that is architecturally separate but one codebase. Every route in the platform is reachable within three taps of the home page.

4.1 Sitemap Overview

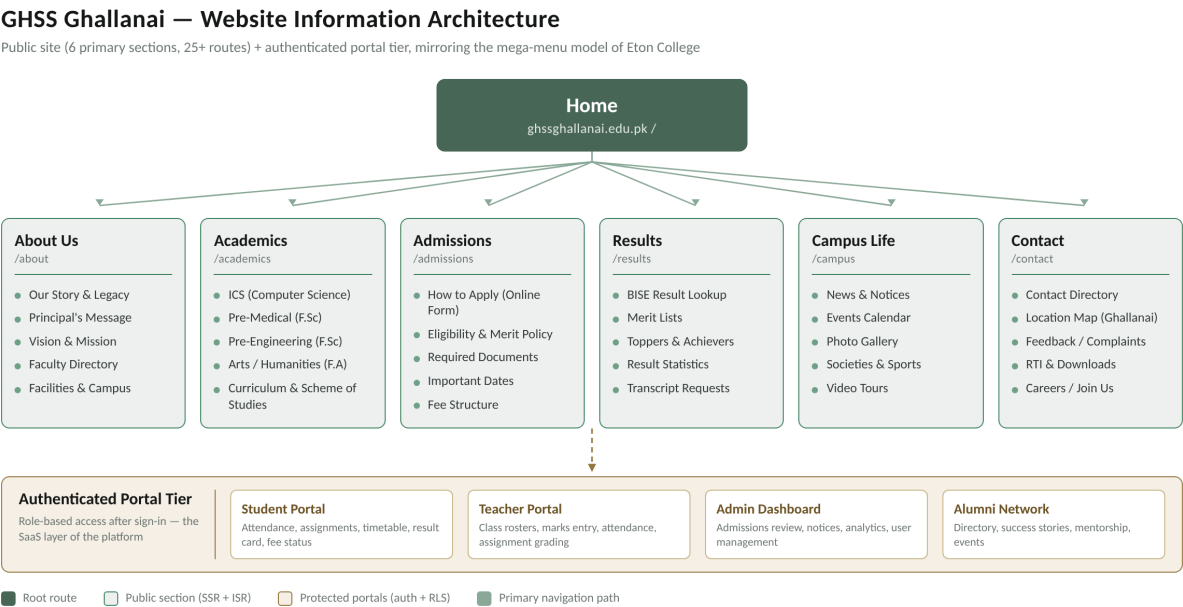


Figure 1: Information architecture: six public sections above a role-based authenticated portal tier.

The public tier deliberately mirrors the navigation grammar of the reference sites. About Us carries the heritage and leadership content that Aitchison and Edwardes use to establish trust. Academics is a hub that expands into the four programme microsites, exactly as TMUC expands its schools of study. Admissions and Results are the two transparency surfaces that no contextual peer combines at this quality level. Campus Life consolidates news, events, gallery, and societies in the PGC tradition. Contact closes the loop with a directory, map, and feedback channel. The portal tier beneath the dashed line is invisible to anonymous visitors in the navigation header, appearing only as a single Sign In action that routes each role to its own workspace.

4.2 Route Registry

Route	Rendering	Auth	Purpose
/	SSR + ISR 60s	Public	Home: hero, ticker, stats, programmes, news, events, testimonials
/about, /about/history, /about/principal, /about/faculty	SSG	Public	Heritage, leadership, vision, faculty directory
/academics, /academics/ics, /academics/pre-medical, /academics/pre-engineering, /academics/arts	SSG	Public	Programme hub and four programme microsites with subject tables
/admissions, /admissions/apply, /admissions/eligibility, /admissions/fees, /admissions/faq	SSG + CSR form	Public (form submits)	Admission guidance and the online application
/results, /results/lookup, /results/merit-list, /results/toppers	SSG + CSR lookup	Public	Result lookup, merit lists, achievements wall
/campus/news, /campus/news/[slug], /campus/events, /campus/gallery, /campus/societies	SSG + ISR 300s	Public	News articles, events calendar, photo albums, societies
/contact, /contact/feedback	SSG + CSR form	Public	Directory, map, feedback and complaint channel
/ur/* (key pages)	SSG	Public	Urdu renderings of home, admissions, fees, results, contact
/portal/* (student, teacher)	CSR	Required	Role workspaces: attendance, assignments, timetable, marks
/admin/*	CSR	Required (admin)	Content management, admissions review, analytics, users
/alumni/*	CSR	Required (alumni)	Directory, stories, mentorship matching
/offline	Static	Public	Service-worker offline fallback page

Table 5: Route registry with rendering strategy and access level for every surface.

4.3 Navigation Model

The desktop header carries the crest logo, six primary items, a search icon, a theme toggle, a language switcher, and a persistent green Apply Now button that follows Exeter's pattern of a permanently visible call to action. Each primary item opens a mega-panel: a two-column dropdown that previews the section's top destinations with a thumbnail and description, so a parent can jump directly to Fee Structure without learning the site's vocabulary. On mobile, the header collapses to logo, theme toggle, and a hamburger that opens a full-screen sheet with accordion sections, large 48-pixel tap targets, and the same Apply Now action pinned to the bottom. Breadcrumbs appear on every nested page, and a command-palette search (activated by the slash key on desktop and a search sheet on mobile) indexes all static routes, news articles, and programme pages.

The footer serves the second navigation grammar: a four-column layout with quick links, contact details including the WhatsApp number, programme shortcuts, and a compact sign-in block for all four portals, mirroring the dual-navigation discipline observed on Eton and TMUC. A thin utility bar above the header announces the current admission status, for example Admissions open for the 2026-27 session, with a link directly into the application, exactly as Aitchison surfaces session announcements at the top of every page.

4.4 Primary User Journeys

1. Ayesha discovers the site through a WhatsApp share, lands on the home hero, taps ICS on the programmes band, reads the subject combination, taps Apply Now, completes the five-step form across two connectivity windows thanks to draft autosave, and receives a WhatsApp confirmation with her application number.
2. Haji Saifullah opens the PWA from his home screen during result week, taps Results, enters his son's roll number, sees the full subject-wise card, and screenshots it for the family group, while a WhatsApp broadcast reaches him first because he opted in at admission time.
3. Mr. Khan signs in during first period, opens the attendance sheet for 1st-year Pre-Engineering, marks four absences with two taps, and the attendance percentage updates instantly for all affected students and their parents' next portal visit.
4. The administrator reviews twelve new applications over chai, promotes nine to the interview list, publishes the merit list with one click that simultaneously updates the website, regenerates the sitemap entry, and triggers the WhatsApp broadcast to all applicants.
5. Naveed registers as an alumnus, is verified against the school's records by the exam branch within a day, appears in the public directory of the class of 2019, and receives a mentorship match request from a Pre-Medical student preparing for MDCAT.

5. Design System: Green and Gold

TOKENS · THEMES · TYPOGRAPHY · ICONS · MOTION · COMPONENTS

The design system is the contract between aesthetics and engineering: every colour, font size, spacing unit, icon, animation, and component on the site is defined once as a token and consumed everywhere. This chapter specifies that system completely, so that any page built from it looks like it belongs to the same institution. The identity pairs the deep green of Pakistan's flag and the Mohmand valleys with the gold of academic laurels and distinction, a combination the research traced across Aitchison's crest-and-foil language and Eton's heritage gilding.

5.1 Brand Identity

The logo is an SVG crest built from three elements: a shield silhouette referencing the academic heraldry that Aitchison, Eton, and Exeter all share, an open book whose pages suggest the four study programmes when counted as leaf pairs, and a rising peak behind the book representing the hills that frame Ghallanai. The crest renders in three locked variants: full colour (emerald shield, gold book, deep-green peak), mono green for stamps and print, and reversed cream for the dark theme and the gold seal version for certificates. A wordmark pairs the crest with the school name set in Playfair Display, with the Urdu name in Noto Nastaliq Urdu beneath it at 60 percent scale. The full lockup, spacing clear-space rules, minimum sizes, and incorrect-usage examples will be delivered as a one-page brand sheet generated from the same SVG sources.

Colour carries the meaning system. Green is institutional: it signals the school, navigation, primary actions, and success. Gold is honour: it highlights merit, achievement, toppers, anniversaries, and the small premium details. Neutrals stay slightly green-tinted rather than pure grey, so the whole surface family feels like one material. Semantic red, amber, and green exist only inside the portal tier for status, and never decorate public pages.

5.2 Colour Tokens

The token architecture uses CSS custom properties defined on the document root and swapped by a data-theme attribute, which is the mechanism that makes dual theming nearly free. The bright theme optimises daylight readability and print; the dark theme reduces glare and OLED battery cost on the phones most students use at night, and follows the 2026 context-aware pattern: it defaults to the operating system preference, remembers a manual override, and never flashes the wrong theme on load.

Token	Bright value	Dark value	Usage
--bg / page background	#FAFDF7	#0B1F14	App canvas
--surface / cards	#FFFFFF	#12291B	Cards, panels, mega-menu
--surface-raised	#F0F7F1	#1A3526	Hover states, table stripes, chips
--primary (green 800)	#14532D	#4ADE80	Primary buttons, links, active nav
--primary-strong (green 900)	#0E3B20	#22C55E	Hero gradient end, footer
--accent (gold 700)	#B8860B	#E9C46A	Badges, toppers, focus rings, key stats
--accent-soft (gold 100)	#F5E6C4	#3A2F14	Announcement ribbons, gold chips
--text-primary	#1A2E22	#E8F5EC	Body text
--text-muted	#5B6B5F	#93A89A	Captions, meta, placeholders
--border	#D7E5D9	#24402E	Hairlines, dividers, inputs
--success / --warning / --danger	#15803D / #B45309 / #B91C1C	#4ADE80 / #FBBF24 / #F87171	Portal status semantics only

Table 6: Green and Gold token table for both themes; every value meets WCAG AA contrast against its background.

Contrast is verified, not assumed. Body text tokens clear 12:1 in bright and 11:1 in dark; gold is never used for running text, only at sizes of 18 pixels and above or on deep-green backgrounds where it exceeds 4.5:1. The --primary token inverts its lightness between themes, going from a deep forest green in bright mode to a luminous emerald in dark mode, which is the single most important trick for keeping one component library coherent across both themes.

5.3 Theme Architecture

Implementation follows the next-themes pattern with Tailwind CSS 4. A single class on the html element (data-theme="bright" or "dark") switches every token; all components reference tokens exclusively, so no component ever contains a theme conditional. An inline script in the document head reads localStorage and the system preference before first paint, eliminating the flash of wrong theme. The toggle itself is a small sun-and-moon animated icon in the header that morphs over 250 milliseconds, with a respect for prefers-reduced-motion honoured by every animation in the system. Theme choice persists per device and, for signed-in users, is also written to their profile so the portals greet them in their chosen theme on any device.

5.4 Typography

The type system pairs Playfair Display for display and headings with Inter for interface and body text, the same serif-plus-modern-sans strategy observed at Aitchison (Playball and IBM Plex Sans) and Eton (serif headlines over a humanist sans), tuned to contemporary rendering. Inter is a variable font, subset to Latin plus basic punctuation at build time, and loaded with font-display swap and a metric-compatible fallback stack so layout never shifts while fonts arrive. Playfair Display loads two weights only, 700 and 900, with a subset restricted to the glyphs actually used in headings. Noto Nastaliq Urdu loads on demand for Urdu pages and Urdu passages inside English pages, with generous line-height because Nastaliq ascends and descends far more than naskh styles.

Role	Font / weight	Size (fluid)	Line height	Notes
Display / hero	Playfair Display 900	clamp(44px, 6vw, 72px)	1.08	Gold accent word per heading allowed
H1 page title	Playfair Display 700	clamp(34px, 4.5vw, 48px)	1.15	With gold underline flourish
H2 section	Inter 700	clamp(24px, 3vw, 32px)	1.22	Kicker eyebrow above in gold caps
H3 subsection	Inter 600	clamp(19px, 2.2vw, 24px)	1.30	
Body large / lead	Inter 400	clamp(17px, 1.6vw, 19px)	1.65	Chapter intros
Body	Inter 400	16px (15px min on small)	1.60	Max 68ch measure
Small / meta	Inter 400	14px	1.50	Dates, captions
Buttons / nav	Inter 600	14-15px	1.0	Letter-spacing 0.01em
Numerals / stats	Inter 600 + tabular-nums	clamp(36px, 5vw, 56px)	1.0	Odometer-ready monospaced digits
Urdu display	Noto Nastaliq Urdu 700	clamp(24px, 3.4vw, 40px)	2.0	RTL direction, right-aligned
Urdu body	Noto Nastaliq Urdu 400	18px	2.2	RTL, on Urdu routes only

Table 7: Type scale: a two-family system with a fluid clamp-based scale and an Urdu track.

5.5 SVG Icon System

Icons are delivered as a single, hand-tuned SVG set on a 24-pixel grid with a 1.75-pixel stroke, round caps, and `currentColor` inheritance, so every icon automatically follows the active theme. The base vocabulary of roughly 45 icons adapts the Lucide open-source set for guaranteed consistency, and the set adds custom school glyphs that no generic library contains: a book-and-flask for Pre-Medical, a compass-and-gear for Pre-Engineering, angle brackets over a book for ICS, a quill-and-palette for Arts, a mosque-and-minaret silhouette for heritage contexts, and a mountain peak mark for Mohmand identity.

Colourful treatment, requested explicitly for feature cards and programme tiles, is achieved with the duotone recipe rather than multi-hue icon fonts: each custom glyph carries a primary path that inherits `currentColor` and a secondary detail path filled with the gold accent at 22 percent opacity. The result is a colourful, modern icon language that still belongs to a single family and costs nothing at runtime. Icons are tree-shaken per page through a build-time sprite: each route imports only its used glyphs, which are inlined as SVG symbols with a `use` element, eliminating the render-blocking icon font entirely. Favicon and PWA icon sets are exported from the same master SVGs at 192 and 512 pixels, including a maskable variant with the crest inside the safe zone.

5.6 Motion and Micro-Interactions

Motion is specified as tokens, exactly like colour. Durations are 150 milliseconds for feedback, 250 for reveals, and 400 for page-level transitions; the house easing is `cubic-bezier(0.22, 1, 0.36, 1)`, a fast-out gentle-in curve that reads as confident rather than bouncy. Scroll-triggered reveals implement the AOS pattern the research admired at TMUC, but with a native `IntersectionObserver` and CSS transitions instead of a jQuery-era plugin: cards fade up 16 pixels with a 60-millisecond stagger inside grids, and once-only semantics so content never re-animates on back-scroll.

- Hero entrance: the headline words rise with a 40-millisecond word stagger, the gold accent word draws its underline via a scaled pseudo-element, and the crest fades in last, completing in 900 milliseconds.
- Statistical odometers: figures count up from zero over 1,200 milliseconds with easing when scrolled into view, using tabular numerals so digits do not jitter, directly reproducing the highest-impact pattern found on TMUC and Exeter.
- Buttons: a 100-millisecond background shift, a 2-pixel lift with soft shadow on hover, and a pressed state that sinks 1 pixel; the primary button also carries a sheen sweep every 6 seconds on the hero only.
- Cards: 4-pixel lift with a green shadow at 12 percent opacity, and a border that transitions to the gold accent when a card is keyboard-focused.
- Mega-menu: panels fade and drop 8 pixels in 180 milliseconds; the mobile sheet slides in from the right with a scrim and a body scroll lock.
- Theme morph: the sun-and-moon toggle cross-fades while token colours transition over 250 milliseconds, giving the whole page a single soft blink rather than a hard swap.

- Route transitions: the View Transitions API fades the main region over 250 milliseconds with a subtle 8-pixel upward drift, falling back to no animation where unsupported.
- Reduced motion: a prefers-reduced-motion media query collapses every duration above 150 milliseconds to instant, and disables odometers and sheens entirely, fulfilling the accessibility commitment of Chapter 11.

5.7 Component Library Inventory

The library ships roughly 28 components in three tiers: primitives (Button in 5 variants, Input, Select, Textarea, Checkbox, Badge, Chip, Avatar, Tooltip, Skeleton), composites (Hero, StatBand with Odometer, ProgrammeCard, NewsCard, EventCard, Timeline, Accordion, Tabs, Breadcrumb, Lightbox, DataTable, NoticeTicker, FeatureGrid, TestimonialCard, FAQAccordion), and overlays (Modal, Drawer, Toast, InstallPrompt, ConfirmDialog). Every component is documented in a living style-guide route inside the admin portal with prop tables, theme previews in both modes, and accessibility notes, so future maintainers extend the system instead of forking it. The style guide is itself a launch deliverable of Phase 0, because it is the mechanism that keeps a single developer consistent across sixteen weeks.

6. Page-by-Page Specifications

EVERY PUBLIC ROUTE · COMPONENT BLOCKS · CONTENT REQUIREMENTS

This chapter specifies each public page as an ordered stack of component blocks with its content requirements and its measure of success. The specifications are deliberately concrete: a builder should be able to construct any page from this chapter without asking a single additional design question.

6.1 Home

The home page is the institution's handshake, and it follows the narrative arc the research rewarded: prove stature, prove programmes, prove life. It assembles, in order: a full-viewport video hero with a dark green gradient scrim, the school name in Playfair Display with a gold accent word, a two-line positioning statement, and two actions (Apply Now primary, Explore Programmes secondary), with the crest watermark and a scroll cue; the notice ticker, a marquee of the three latest notices that pauses on hover and links to each; the statistics band, four odometer counters (students enrolled, board positions, scholarship holders, alumni) on a deep-green panel with gold figures; the programmes band, four colourful duotone cards for ICS, Pre-Medical, Pre-Engineering, and Arts, each with icon, one-line promise, subject count, and a chevron into its microsite; the why-us band, three feature cards with gold duotone icons covering faculty, facilities, and results transparency; the news and events band, a two-column split of the three latest news cards and a compact events list with date chips; the voices band, a testimonial carousel of student and alumni quotes with names, classes, and photographs; and the CTA band, a gold-on-green closing panel with the admission status, deadline countdown, and the WhatsApp click-to-chat button adopted from TMUC.

The hero video is a 12-second silent loop of campus life, delivered at 720p in AVIF-derived poster plus MP4 and WebM sources, preloading only the poster on mobile connections and muting autoplay for LCP safety. Home content is ISR-revalidated every 60 seconds so the ticker and news band reflect the admin dashboard within a minute of publishing, which is the mechanism behind the under-five-minutes notice KPI in Chapter 1.

6.2 About Section

About Us opens with the school's story told as a vertical timeline with dated entries, vintage photographs where available, and a gold marker on milestones, following the heritage presentation of Aitchison and Edwardes. The Principal's Message page pairs a formal portrait with a 300-word message in both English and Nastaliq Urdu in side-by-side panels, plus signature and tenure. Vision and Mission render as two cards over the crest watermark. The Faculty Directory lists every teacher as a card with photograph, name, qualification, subject specialisms, and years of service, filterable by department, with a print stylesheet so the office can produce the wall poster version. Facilities documents classrooms, science laboratories, the computer lab, library, grounds, and any hostel provision, each with a photo gallery and capacity facts, which is content the admissions journey links into directly.

6.3 Academics and the Four Programme Microsites

The Academics hub presents the four programmes as large cards with intake numbers and subject counts, above a scheme-of-studies explanation and the academic calendar. Each programme microsite follows one identical template, which is what keeps four pages maintainable by one developer: a hero with the programme's duotone icon and a one-line promise; an overview of three paragraphs; the subject combination table for first and second year with marks weights; entry requirements including required matric subjects and minimum percentages; the careers pathway section naming the professions and university programmes the stream feeds, with alumni examples where available; the assessment pattern explaining BISE annual examinations, practicals where applicable, and internal test weight; and a closing CTA into admissions with the eligibility table link. The ICS page additionally documents the computer lab and programming languages taught; Pre-Medical documents the biology laboratory and MDCAT guidance; Pre-Engineering documents mathematics clubs and ECAT preparation; and the Arts page documents the humanities elective combinations and career paths in law, civil service, and media, countering the stream stigma the research team noted in the district.

6.4 Admissions

The admissions section is engineered around the parent's decision sequence: understand, prepare, apply, track. The landing page presents a four-step visual journey (Check Eligibility, Prepare Documents, Apply Online, Track Status) as an interactive timeline, with the current session's dates table (opening, deadline, test, interview, merit list, enrolment) and a downloadable checklist PDF. The Eligibility page carries the requirements table per programme with minimum matric marks and required subject combinations. The Fee Structure page presents the official fee table, any scholarship and concession policies, and the bank details, with a note that online payment arrives in v2 and a WhatsApp contact for fee questions, matching PGC's parent-first financial transparency. The FAQ page is an accordion of twenty questions gathered from office staff, marked up with FAQPage structured data for search rich results. The Apply page hosts the five-step application form specified in Chapter 7, with draft autosave, document upload, and the confirmation screen that issues the application number and triggers the WhatsApp receipt.

6.5 Results

The Results section is the transparency flagship. The lookup page offers a single prominent form (roll number, select year and programme, optional captcha after three failed attempts) that renders the full subject-wise result card with a printable layout and share buttons. The Merit Lists page publishes each session's lists as accessible HTML tables (not scanned images) with PDF downloads, dated and versioned, plus an archive of previous years. The Toppers wall presents position holders as honour cards with gold styling, photographs, and short interviews, feeding the same content into the home statistics band. A Result Statistics page charts the school's five-year performance trend by programme in simple, high-contrast charts, giving the district evidence of institutional improvement; all result data flows from the supervised import pipeline specified in Chapter 7.

6.6 Campus Life

Campus Life consolidates the living institution. News runs as a dated list with category chips (Academic, Achievement, Notice, Event, Sports) and article pages with a reading-time estimate, share actions, and related links, marked up with Article structured data. The Events calendar offers month and list views with event pages carrying date, venue, and poster images, marked up with Event structured data so occasions such as the annual prize distribution can surface in search. The Gallery is a masonry grid of albums (campus, ceremonies, sports, Pashto Culture Day) opening into a swipeable, keyboard-navigable lightbox with captions, fed by the Cloudinary pipeline. The Societies page documents each club and society with its patron teacher and meeting cadence, seeded with content from staff interviews during Phase 1.

6.7 Contact

The Contact page centres an embedded map pinning the campus in Ghallanai with written directions from Mohmand's main junctions, a contact directory table (office, exam branch, admissions, principal) with phone numbers and office hours, the WhatsApp business number as a prominent click-to-chat card, and a feedback and complaint form that files into the admin dashboard with an acknowledgment message, following the complaint-channel pattern observed at Islamia College. A Downloads block offers the prospectus, admission forms, and fee documents, each versioned and dated. The page closes with visiting hours and a note on required documentation for visitors, which the office staff flagged as a real pain point during research.

7. Feature Modules

ADMISSIONS · RESULTS · PORTALS AND LMS · ALUMNI · ADMIN · NOTIFICATIONS

The four feature modules selected for this platform, online admissions, the results system, the student and teacher portals with an LMS core, and the alumni network, plus the admin dashboard and notification engine that serve them, are specified here to field level. Each module lists its data entities, its user flows, and its acceptance criteria, so that Chapter 12's weekly milestones reference unambiguous definitions of done.

7.1 Online Admissions

The admissions module replaces the paper first-information sheet with a five-step guided form: personal details (name, father's name, CNIC/B-form numbers, contact, WhatsApp opt-in), academic history (matric board, roll number, obtained and total marks, year, previous school), programme selection (one of the four streams with automatic eligibility pre-check against matric marks and subjects), document upload (photograph, B-form, matric result card, domicile, and any concession proof, stored in Supabase Storage), and review-and-submit with a declaration checkbox. Drafts autosave after each step to survive connectivity drops, the form validates inline with Urdu error messages available, and submission issues a formatted application number, an email receipt, and a WhatsApp confirmation carrying the number and next steps.

The admin review queue lists applications filterable by programme, status, and date, showing eligibility flags computed from the submitted marks. Reviewers promote applications through statuses (received, under review, shortlisted for test or interview, offered, admitted, or rejected with reason), record interview notes, and bulk-publish merit lists: the publish action generates the ranked list from admitted-mark applications against seat quotas, renders it as accessible HTML and PDF, posts it to the merit-list page, and triggers the WhatsApp broadcast. Every status change writes to an audit table, satisfying the accountability expectation of a government institution.

ACCEPTANCE CRITERIA

A first-time applicant on a low-cost Android phone completes the form within 15 minutes across at least one dropped-connection window without losing entered data; an administrator publishes a merit list and 300 WhatsApp notifications dispatch within 5 minutes of a single click; no application is ever editable after submission except by an admin action that is logged.

7.2 Results and Merit Lists

The results module ingests data through a supervised import: the exam branch downloads the official BISE result file, a CLI script validates and imports it into the marks tables with a checksum report, and two staff members confirm counts before publication, replacing fragile manual entry. The public lookup renders the subject-wise card with grades, marks, and the school's own class-based position where applicable, and the same data powers the toppers wall and the statistics charts. Internal assessment flows from the teacher portal into the same schema, so a student's portal shows board results and internal test history side by side. Rate limiting and a captcha after repeated failures protect the lookup from scraping, and every published result page ships with the year and programme in its title for search visibility.

7.3 Student and Teacher Portals with LMS Core

The portals are the daily workspace of the school. The student portal opens onto a dashboard of today's timetable, unread notices, pending assignments, and an attendance percentage ring; deeper tabs hold assignments (due, submitted, graded with teacher remarks), the downloadable resource library organised by programme and subject, the personal result history, and the fee status view. The teacher portal mirrors it: class rosters with photographs, one-tap attendance marking with offline queuing for connectivity drops, assignment creation with due dates and attachments, a gradebook grid with CSV export, and a leave-application workflow into the admin queue. All portal routes are client-rendered behind Supabase Auth with row-level security scoping every query to the signed-in identity, per the architecture in Chapter 8.

LMS capability at launch	Student experience	Teacher experience
Assignments	View brief, attach submission (photo or PDF), see grade and remarks	Create with attachment, set due date, download submissions, grade with remarks
Resources	Browse subject library, read PDFs in-app, download for offline	Upload and categorise notes, past papers, and syllabi
Attendance	See daily log and percentage per subject	Mark a class in under a minute, view history, export monthly report
Notices	Receive targeted notices with read receipts	Draft notices for approval by admin
Timetable	Personalised weekly view	Class-wise and teacher-wise views
Grades	Internal test history and board results in one card	Gradebook grid with auto-computed aggregates

Table 8: Launch-scope LMS capability matrix for the student and teacher portals.

7.4 Alumni Network

The alumni module follows the Aitchison Connect model at public-school scale. Registration collects name, roll number, batch years, programme, current occupation, organisation, city, and contact preferences, and verification is human: the exam branch matches the record against admission registers, typically within one working day, before the profile activates. Verified alumni appear in a searchable, filterable directory (by batch, programme, city, and field), can publish a success story with photographs, and can volunteer as mentors; senior students then request mentorship matches filtered by field, and the platform brokers the WhatsApp introduction rather than exposing contact details directly. An annual reunion event page and a class-notes feed complete the launch scope, with every story feeding the home voices band through the same content pipeline as news.

7.5 Admin Dashboard

The admin dashboard is the SaaS control room: a sidebar application with content management (news, notices, events, gallery albums, faculty, and FAQs as guided forms with live preview), the admissions review queue, results publishing, user and role management, and an analytics view of the platform's own telemetry (visitors, top pages, admission funnel conversion, PWA installs, and WhatsApp opt-ins, sourced from Vercel Analytics plus internal events). Every destructive action requires a typed confirmation, every mutation is audit-logged, and a soft-delete archive keeps a 90-day recovery window, because a school cannot afford a bad Tuesday morning. The dashboard ships in dark and bright themes from the same token system, and its forms carry Urdu microcopy for office staff who are more comfortable in Urdu.

7.6 Notification Engine

Notifications are the connective tissue between modules and the community's actual communication habit. The engine defines triggers and channels declaratively: admission receipt and status changes, merit list publication, result publication, new notice, assignment posting, and attendance threshold alerts for parents. Delivery fans out through the WhatsApp Business API for time-critical alerts (templated messages, pre-approved), transactional email through Resend for receipts and digests, in-app notifications for portal users, and web push for installed PWAs. Opt-in is explicit per channel at admission and inside the portal, every message carries an unsubscribe instruction, and a quiet-hours rule (no sends between 21:00 and 07:00 except emergencies) respects family life, a policy adopted from the notification etiquette observed across the reference sites.

Trigger	WhatsApp	Email	In-app	Web push
Application received / status change	Yes	Yes	-	-
Merit list published	Broadcast	Digest	Yes	Yes
Result published	Broadcast	Digest	Yes	Yes
New notice	Broadcast	-	Yes	Yes
Assignment posted / graded	-	-	Yes	Optional
Attendance below 75% (parent alert)	Yes	Yes	-	-
Fee reminder (v1: manual schedule)	Yes	Yes	-	-

Table 9: Notification trigger matrix across the four delivery channels.

8. Technology Architecture

STACK RATIONALE · RENDERING · DATA MODEL · AUTH · MEDIA · APIS

The architecture optimises for one decisive constraint identified in Chapter 3: a single motivated student developer must be able to build, deploy, and alone maintain a platform that feels like a software product. Every choice below trades operational cleverness for managed reliability, and every service runs on a free tier at launch scale without watermarks or expiries.

8.1 Stack Rationale

Layer	Choice	Why this and not the alternative
Framework	Next.js 16, App Router, TypeScript	One codebase for public site, portals, and admin; server components remove most client-side loading waterfalls; the App Router's file semantics are the most documented path for a solo learner; TypeScript catches an entire class of silent data bugs
Styling	Tailwind CSS 4 with CSS-variable tokens	Token architecture for dual themes; zero unused CSS at build; visual decisions stay in markup where one developer works anyway
UI primitives	shadcn/ui (Radix-based), customised	Accessible composites (dialog, menu, tabs) would take weeks to hand-build correctly; shadcn's copy-in model keeps them editable rather than a black-box dependency
Database	Supabase PostgreSQL	Managed Postgres with row-level security, realtime, storage, and auth in one free project; alternatives require self-hosting and backups the developer cannot guarantee
ORM	Prisma	Typed schema as documentation; migrations as reviewable files; the schema below is expressed directly in Prisma terms
Media	Cloudinary	Automatic AVIF/WebP, responsive breakpoints, on-the-fly transforms, and a generous free tier; keeps image optimisation off the Vercel image budget and off the developer's plate
Hosting	Vercel	Preview deployment per pull request is the solo developer's safety net; edge CDN for Pakistani audiences; zero-config Next.js support
Source control / CI	GitHub + Actions	Free CI minutes run lint, type-check, tests, and Lighthouse gates before any deploy
Email	Resend	3000 free transactional emails monthly with a clean API for receipts and digests
Messaging	WhatsApp Business Cloud API	The community's actual channel; per-message cost is small and scoped to critical alerts only
Analytics	Vercel Analytics + Microsoft Clarity	Privacy-light traffic telemetry plus free session heatmaps, the exact stack observed on Eton

Table 10: Technology choices with explicit rationale against the solo-maintainer constraint.

8.2 System Architecture

GHSS Ghallanai — SaaS Technology Architecture

Next.js 16 on Vercel, Supabase data platform, Cloudinary media pipeline, GitHub CI/CD — 100% free-tier at launch

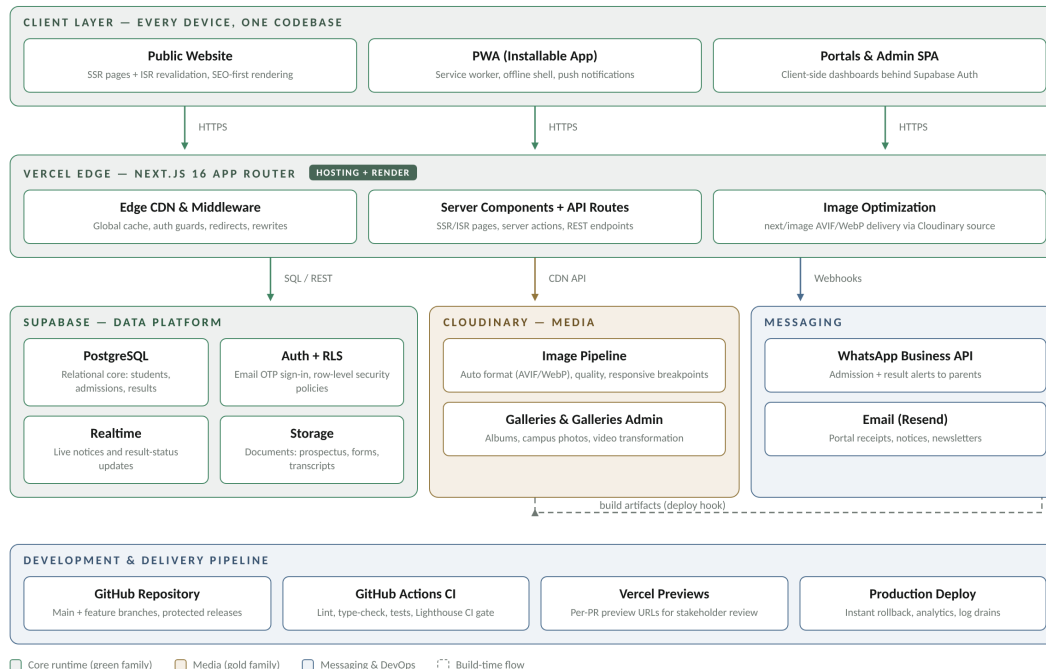


Figure 2: System architecture: client tier, Vercel edge rendering, Supabase data platform, Cloudinary media, messaging services, and the GitHub delivery pipeline.

The request path is deliberately boring. Anonymous visitors hit Vercel's edge CDN, which serves server-rendered pages and revalidates them on the schedule fixed in the route registry; images resolve through the Cloudinary loader with next/image so layout is stable and formats negotiate automatically. Signed-in traffic exchanges a Supabase session, after which every portal query passes PostgreSQL row-level security, meaning a student's request can only ever read rows the policy admits, even if the application code were mistaken. Write paths from forms flow through server actions that validate input, enforce rate limits, and call the notification engine. The delivery pipeline keeps the developer honest: every change lands through a pull request, and a pull request is not mergeable until type-checks, tests, and the Lighthouse CI gates pass, after which Vercel promotes the build to production behind instant rollback.

8.3 Rendering Strategy

- Static generation with ISR for programme pages, about pages, and reference content: build once, revalidate on demand after edits.
- ISR 60 seconds for the home page and news list; ISR 300 seconds for events and gallery listings; article pages revalidate on publish webhooks.
- Server-rendered lookup results with caching disabled per request, so result and merit data is always fresh.

- Client-side rendering inside all portals and admin, with optimistic updates and skeleton loaders, because dashboards are stateful workspaces, not documents.
- Streaming with Suspense boundaries on the home page so the hero paints before the news band's data resolves, protecting the LCP budget.

8.4 Repository Layout

```
ghss-digital-campus/  
  app/  
    (public)/          # marketing site route group  
      page.tsx about/ academics/[programme]/  
      admissions/ results/ campus/ contact/ ur/  
    (portal)/          # authed route group (layout enforces session)  
      student/ teacher/  
    (admin)/admin/      # role-guarded admin shell  
    (alumni)/alumni/  
    api/               # webhooks: cloudinary, whatsapp status, cron  
  components/ ui/ icons/ # design system + shadcn primitives  
  lib/ db/ auth/ notify/ # prisma client, RLS helpers, engine  
  content/             # editorial markdown for static pages  
  public/ manifest.json sw/  
  supabase/  
    migrations/ policies/ functions/ seed/
```

8.5 Data Model

The schema is grouped by domain and totals nineteen tables, every one carrying `created_at` and `updated_at` timestamps and soft-delete flags where history matters. The tables below give the column-level contract for the core; auxiliary tables (tags, settings, audit_log, files, and the FAQ store) follow the same conventions.

Table	Key columns	Relationships and notes
profiles	id (uuid, FK auth.users), full_name, phone, whatsapp_opt_in, role enum(student, teacher, admin, alumni), theme_pref, avatar_url	One row per authenticated human; role gates every portal route
classes / sections	class_id, year enum(1st, 2nd), programme enum, section_label, class_teacher_id	Sections are the teaching unit; rosters join students to sections
students	student_id, profile_id?, admission_no, roll_no, section_id, guardian_name, guardian_phone, status enum	Links to admissions.admission_no on enrolment
subjects / teacher_assignments	subject_id, title, code, programme, year / teacher_id, subject_id, section_id, session_year	The teaching timetable is derived from assignments
attendance	id, section_id, subject_id, date, student_id, status enum(p, a, l, leave), marked_by	Composite-indexed on (section, date); parent alerts query this
assignments / submissions	assignment_id, section_id, subject_id, title, brief, attachment_url, due_at / submission_id, assignment_id, student_id, file_url, grade, remarks, graded_by	Submissions store Cloudinary URLs; grading writes notification events
marks / board_results	exam_id, student_id, subject_id, total, obtained, grade / year, programme, roll_no, subject-wise columns, import_batch_id	Board results arrive by supervised import with checksum audit
admissions / admission_docs	application_no, programme, personal + academic fields, status enum, merit_rank, decision_note / doc_type, file_url	Status machine drives the review queue and WhatsApp triggers
notices / events / news_posts	title, body, audience enum, published_at, pinned / event fields / slug, category, cover_url, body_md	All publish flows go through admin approval; realtime-capable
gallery_albums / gallery_photos	album_id, title, cover_url / photo_id, album_id, cloudinary_id, caption, sort	Admin uploads via signed widget; transformations on delivery
alumni_profiles / mentorships	alumni_id, batch_year, programme, occupation, org, city, verified / mentor_id, student_id, status, notes	Verification flag set manually by exam branch

Table 11: Core relational schema by domain; all nineteen tables share timestamp and audit conventions.

8.6 Authentication and Authorisation

Authentication is Supabase Auth with email OTP as the primary factor: office-issued institutional email for staff, personal email plus guardian phone for students, and email for alumni, with passwords optional as a second factor rather than the only secret. Authorisation is layered: Next.js middleware checks the session cookie and role claim before a portal route renders, layout-level guards keep role boundaries in one auditable place, and PostgreSQL row-level security provides the final, code-independent defence. Representative policies: students can select rows where their `student_id` equals auth-mapped identity across attendance, marks, and submissions; teachers can read their assigned sections and write attendance and marks only for those sections; admins bypass nothing implicitly, and instead hold an explicit policy set that the audit table records. Password resets, session revocation, and the annual leavers' de-provisioning are all documented runbooks in the handover pack of Chapter 15.

8.7 Media Pipeline

All photography flows through one pipeline. Admin-side, a signed Cloudinary upload widget drops originals into a raw folder with moderate face-blurring available for consent-sensitive contexts. Delivery-side, the next/image loader requests AVIF or WebP at the exact rendered width from a defined set of breakpoints, applies an auto-quality floor, and carries a tiny blurred LQIP placeholder generated at upload time. The gallery lightbox progressively loads higher resolutions; the hero video ships as three sources with a poster and is paused entirely under prefers-reduced-motion and on metered connections detected through the Network Information API where available. Consent discipline is a school-specific requirement: photographs of students publish only where a guardian media-consent flag, captured at admission, is true, and the flag is honoured automatically by the pipeline.

8.8 API Surface

Surface	Mechanism	Consumers
Public content reads	Server components + Prisma (no public REST)	SSR pages
Form submissions	Server actions with zod validation + rate limit	Admissions, feedback, alumni registration
Result lookup	Route handler with IP rate limit + captcha escalation	Public results page
Cloudinary webhook	/api/webhooks/cloudinary (signature-verified)	Media metadata sync
WhatsApp delivery status	/api/webhooks/whatsapp	Notification engine receipts
Scheduled jobs	Supabase cron functions: quiet-hour queue flush, sitemap ping, backup check	Ops
Realtime channels	Supabase realtime on notices and admission status	Portals

Table 12: API surface: server-first, with only four narrow HTTP endpoints exposed.

9. PWA Strategy

SERVICE WORKER • OFFLINE • INSTALLABILITY • PUSH • APP FEEL

The progressive web application is the platform's delivery answer to Mohmand's reality: expensive data, intermittent coverage, and phones with limited storage. A PWA gives the school an installable, offline-tolerant, push-capable presence without app-store bureaucracy, and the research found no peer institution, in Pakistan or abroad, that has shipped a full one. This chapter specifies the strategy to the file level.

9.1 Service Worker Plan

Caching is generated by Serwist (the maintained successor to next-pwa) at build time, with three explicit strategies. The application shell, layout CSS, the icon sprite, the two web fonts, and the offline page are precached on install, so a cold start on a dead network still opens a branded shell. Public pages follow stale-while-revalidate with a network timeout of 4 seconds: a returning visitor on weak coverage sees the last good copy instantly while the update arrives in the background, which is precisely the behaviour a results-hungry parent needs. Cloudinary images use cache-first with a 30-day expiry and LQIP placeholders, and API route handlers are never cached. The update flow is user-respecting: a new service worker activates in the background, and the next navigation reveals a small toast, A new version is available, with a Refresh action, rather than silently swapping content mid-read.

9.2 Manifest and Installability

Manifest field	Value
name / short_name	GHSS Ghallanai / GHSS Ghallanai
start_url / scope	/?source=pwa / /
display / orientation	standalone / portrait-primary
theme_color / background_color	#14532D (bright) / #FAFDF7; dark manifest variant #0B1F14 via media query asset
icons	192 and 512 PNG plus a maskable variant with the crest inside the 80% safe zone; SVG-any icon for modern browsers
shortcuts	Result Lookup (/results/lookup), Apply Now (/admissions/apply), Notices (/campus/news?cat=notice)
screenshots (install UI)	Home, Result card, Notice list at 1080x1920 for richer install prompts
categories / lang / dir	education / en, ur / ltr (ur routes declare rtl)
share_target	Accepts shared links into the feedback form (v1.1)

Table 13: Web app manifest specification for installability across Android, KaiOS-era browsers, and desktop.

Install prompts are earned rather than shouted. A custom InstallPrompt card appears only after the Chrome engagement heuristics fire (typically a second visit on a second day), offering Add GHSS to your home screen with a short Urdu line explaining the benefit, and the prompt permanently dismisses on decline. On iOS Safari, where the generic share-menu flow persists, a one-time guided tooltip points to the Add to Home Screen action, and the standalone splash uses the manifest icon over the brand green so the launch moment feels owned.

9.3 Push Notifications

Web push is offered at PWA install and at WhatsApp opt-in moments, and it reuses the same declarative trigger matrix as Chapter 7, so a notification is authored once and fanned out to every opted-in channel. Subscription keys are stored against profiles with per-topic flags (results, admissions, notices) that the user controls from a small notification preferences sheet. Because 21:00-to-07:00 quiet hours apply, the Supabase cron queue holds night-time messages for a 07:00 morning flush, except the emergency flag reserved for genuine urgency, a policy the principal's office reviews each term.

9.4 App Feel

- Standalone display with theme-coloured status bars in both themes, and safe-area insets respected on notched devices.
- Bottom navigation bar on mobile inside the portals (Home, Timetable, Assignments, Notices, Profile), matching the grammar of the apps students already know.

- Route transitions through the View Transitions API with the reduced-motion collapse from Chapter 5.
- Optimistic UI everywhere in portals: attendance taps, assignment submissions, and notice reads apply instantly and reconcile silently.
- An offline badge surfaces honestly when the network is gone, and queued writes flush with a small success toast when coverage returns.

10. SEO and Digital Growth

TECHNICAL SEO · STRUCTURED DATA · LOCAL SEARCH · CONTENT ENGINE

Search is how a family in Mohmand will first meet the school, usually on a phone, usually through a query typed in English or a voice query spoken in Urdu or Pashto. The SEO strategy is therefore technical-first (rendering, metadata, structured data, speed budgets set in Chapter 11) and content-second (a publication engine that keeps the site alive), with local search treated as its own front because the school's true competition for the next three years is not another website but word-of-mouth alone.

10.1 Technical Foundation

- Every route exports complete metadata (title under 60 characters, description under 155, canonical, Open Graph, and Twitter cards) through the Next.js metadata API, with per-programme titles such as ICS Admission in Ghallanai - GHSS Ghallanai.
- Dynamic sitemap.xml generated from the database at request time with ISR, covering static routes, news articles, events, programmes, and merit-list pages, plus sitemap entries for the Urdu key pages with hreflang alternates.
- robots.txt allowing all public routes, disallowing portal and admin prefixes, and referencing the sitemap; a Google Search Console property and a Bing Webmaster property verified at Phase 1 and monitored weekly.
- OG images generated through Cloudinary text overlays with the crest, the page title, and the brand green, so shared links render like institutional cards in WhatsApp and Facebook, which is where the district actually shares links.
- Clean URL grammar from the route registry (lowercase, hyphenated, no parameters), 301 redirects mapped from day one, and trailing-slash canonicalisation fixed in next.config.

10.2 Structured Data

The site ships five JSON-LD vocabularies, each bound to the page type that earns it: EducationalOrganization and School on the home page (name, address in Ghallanai, geo coordinates, telephone, sameAs social profiles, and a SchoolChReport card once data exists); Course on each of the four programme pages (programme name, description, provider, and educational alignment to the BISE intermediate level); BreadcrumbList on every nested route; FAQPage on the admissions FAQ; and Event on event pages with dates and venues. Rich results are not guaranteed, but the markup costs one React component and materially changes how the school's snippets look for exactly the queries families type, and the implementation is a single afternoon of Phase 1, not a project of its own.

```
{
  "@context": "https://schema.org",
  "@type": "School",
  "name": "Government Higher Secondary School Ghallanai",
  "alternateName": "GHSS Ghallanai",
  "address": { "@type": "PostalAddress",
    "addressLocality": "Ghallanai",
    "addressRegion": "Khyber Pakhtunkhwa", "addressCountry": "PK" },
  "telephone": "+92-XXX-XXXXXXX",
  "identifier": { "@type": "PropertyValue",
    "name": "EMIS", "value": "filled-from-office-records" }
}
```

10.3 Performance as an SEO Input

Core Web Vital	Budget (4G, mid-range Android)	Primary engineering lever
LCP	2.0 s at launch, 1.6 s by month 6	Hero poster-first, font subsetting, edge CDN, streaming hero
INP	200 ms or better	Server components, minimal hydration on public pages
CLS	0.05 or better	Fixed aspect boxes, LQIP placeholders, font metric fallbacks
TTFB (regional)	Under 400 ms	Vercel edge regions plus ISR; no cold serverless on public paths
JS transferred (public)	Under 120 KB gzip	Almost no client components outside the header and galleries

Table 14: Core Web Vitals budgets and the engineering decisions that meet them.

10.4 Local Search Strategy

The school's Google Business Profile is claimed, verified, and completed in Phase 1 with photographs, office hours, the WhatsApp number, and programme keywords in the description, because the Map pack is the first result for navigation-style queries in the district. Name-address-phone consistency is enforced across the site footer, the profile, and every future listing. The content engine then targets the queries families actually formulate, with one authoritative page per intent rather than scattered posts.

Keyword family	Example queries	Target page
Brand	ghss ghallanai; ghss ghallanai admission; ghss ghallanai result	Home / Admissions / Results
Programme + place	ics admission mohmand; fsc pre medical ghallanai; fa arts college kpk	Programme microsites
Board intent	bise result ghallanai roll number; merit list ghallanai 2026	Results lookup / Merit lists
Parent intent	best government college mohmand; ghallanai college fee structure	Home / Fee structure
Career guidance	mdcat preparation from mohmand; ecat or ics after matric	Programme career sections + guides

Table 15: Local keyword families and the single authoritative page each targets.

10.5 Content Engine and Social Loop

The first-year editorial calendar commits to one substantial piece per week in term time: alternate weeks between achievement stories (topper interviews with photographs, verified before publication), guidance explainers written with subject teachers (subject selection guides, MDCAT and ECAT primers, scholarship explainers), and institutional news (ceremonies, society events, facilities upgrades), with every story cross-posted as a WhatsApp broadcast and a Facebook post that links back to the article, because the district's discovery graph runs through those two channels, not through search alone. Result days receive special editorial treatment: a same-day toppers story, a statistics chart image sized for WhatsApp sharing, and next-day guidance for the following cohort, converting the school's highest-attention moments into its strongest growth moments.

11. Performance, Accessibility and Security

BUDGETS • WCAG 2.2 AA • HARDENING • PRIVACY

These three disciplines are grouped because together they define the floor of quality beneath every feature chapter: the site must be fast on the devices people actually own, usable by people of every ability, and safe with the data families entrust to a government institution. None of them is a final polish phase; each is enforced continuously by the CI gates defined here and scheduled in Chapter 12.

11.1 Performance Discipline

The performance method is budget-driven, not hope-driven. The Core Web Vitals budgets of Chapter 10 are asserted in Lighthouse CI on every pull request against a scripted mid-range Android profile, and a regression beyond budget fails the build exactly as a failing test would. The discipline that keeps the budgets honest is front-loaded: server components render almost the entire public site, so the client JavaScript shipped to anonymous visitors stays under 120 KB gzipped; images always declare intrinsic sizes and load through the responsive pipeline; web fonts are subset at build; and any third-party script, including analytics, must be added to a written allowlist with a documented reason, because every kilobyte of third-party JavaScript is the most expensive kilobyte on the page.

11.2 Accessibility (WCAG 2.2 AA)

Requirement	Implementation standard
Contrast (both themes)	Automated check in CI on the token table; body and UI text at 4.5:1 minimum, large display at 3:1
Keyboard operability	Visible gold focus ring on every interactive element; skip-to-content link; mega-menu and lightbox fully keyboard-navigable with Escape handling
Screen-reader semantics	One h1 per page and ordered headings; landmark regions; aria-live for ticker and form receipts; every icon-only control carries an accessible name
Form accessibility	Programmatic labels, inline error messages tied to inputs, Urdu error text available, and no reliance on colour alone
Motion and vestibular safety	prefers-reduced-motion honoured by every animation token; no autoplaying motion beyond the muted hero video, which is paused for reduced-motion users
Target sizes (2.2)	Minimum 24x24 px hit areas; primary navigation and portal bars at 44-48 px
Focus appearance (2.2)	Focus indicator at least 2 px thick with 3:1 contrast, satisfied by the gold ring
Urdu RTL correctness	dir attributes on Urdu passages, logical CSS properties throughout, mirrored icons audited

Table 16: Accessibility launch gate mapped to WCAG 2.2 AA success criteria.

11.3 Security Hardening

- Defence in depth on data: role checks in middleware and layouts, and row-level security in PostgreSQL as the independently enforced final layer, so a single application bug cannot leak another student's marks.
- Input validation with zod at every server action and route handler, including file type and size checks on uploads, and output escaping by default through React's rendering model.
- Rate limiting per IP and per identity on the result lookup, application submission, feedback, and login OTP endpoints, with exponential backoff and captcha escalation on abuse.
- Content Security Policy with nonce-based script allowances, frame-ancestors none on portal routes, and a locked-down referrer policy, configured through next.config headers.
- Secrets exclusively in Vercel and Supabase environment stores, never in the repository; the one accidental-commit rule is a same-day rotation runbook.
- Session hygiene: OTP throttling, session revocation from the admin user manager, and automatic expiry windows appropriate to shared family devices.
- Backups: daily automated Supabase database backups to a separate storage bucket with a documented, actually rehearsed restore procedure each term.
- Audit trail: every administrative mutation (publish, status change, deletion, role grant) writes an immutable audit row with actor, target, and timestamp.

11.4 Data Privacy

The platform collects exactly what the school's processes require: identity and guardian contact data for students, application data for admission candidates, and voluntary career data for alumni. The privacy posture is documented in plain language on the site (in English and Urdu): what is collected, why, who in the institution can see it, how long admission records of unsuccessful candidates are retained (two sessions), and how a parent may request correction or deletion through the office. Student photographs publish only under the guardian consent flag captured at admission and honoured automatically by the media pipeline. Analytics is aggregate and cookie-light by design, with no cross-site advertising trackers, which is both an ethical choice for a school audience and consistent with the data-minimal posture of the public sector.

12. Development Roadmap

SIXTEEN WEEKS · FIVE PHASES · FOUR MILESTONES

The roadmap converts every specification in this plan into a sixteen-week, single-developer execution schedule with four milestone gates. Phases deliberately overlap where one phase's polish is another's setup, and every phase ends with something demonstrable to the principal's office, because sustained institutional support is itself a project resource that must be fed on schedule.

12.1 Timeline Overview

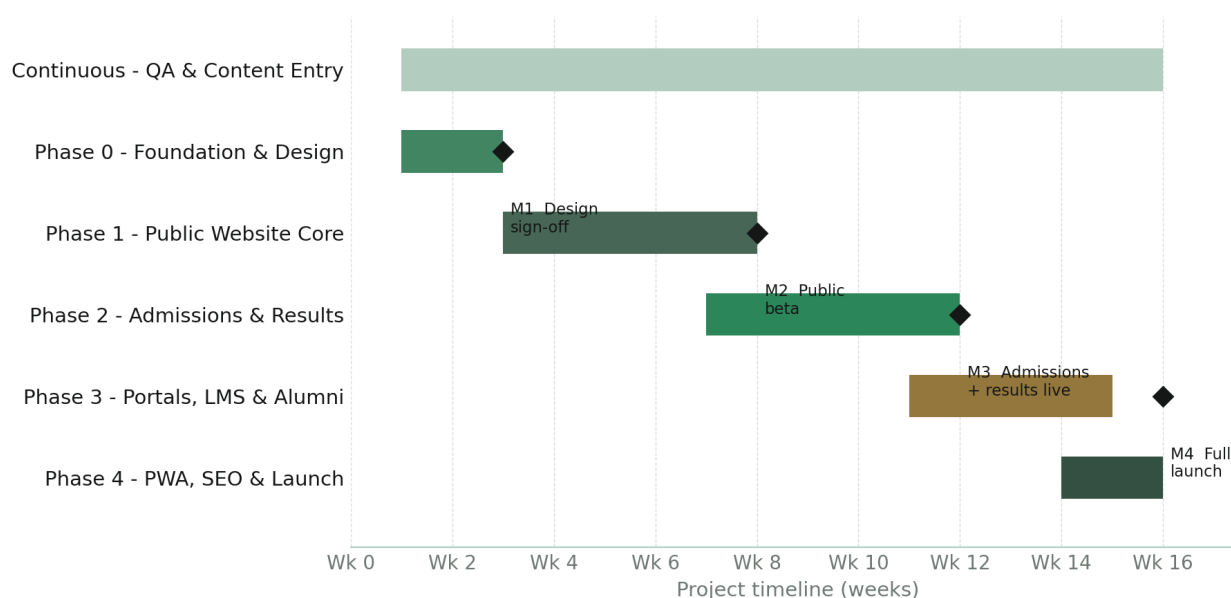


Figure 3: Sixteen-week roadmap: five phases with overlapping hand-offs and four milestone gates.

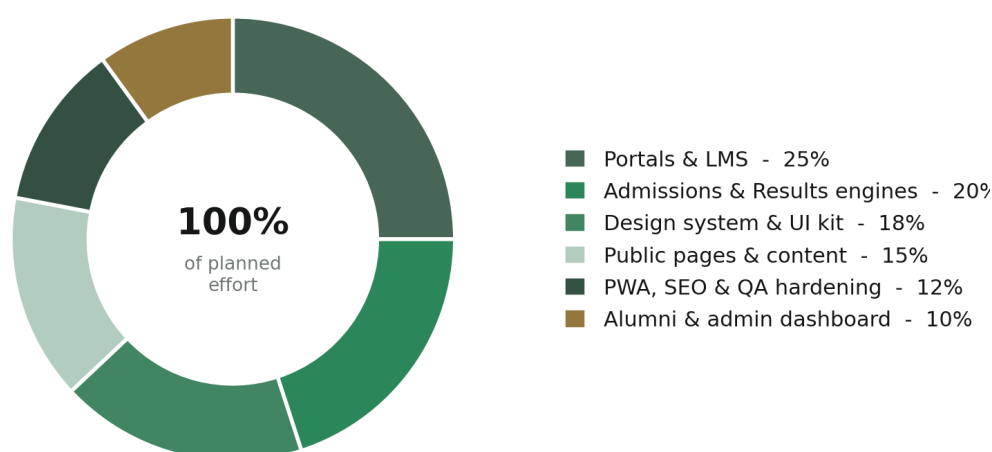


Figure 4: Planned effort distribution across workstreams; portals and admissions engines carry the largest shares.

12.2 Phase Specifications

Phase	Weeks	Deliverables	Exit criteria (milestone gate)
Phase 0: Foundation and design system	1-3	Repository, Supabase project, Prisma schema v1, design tokens and component library, style-guide route, logo and icon set, wireframes for all templates	M1: design system signed off; every core component renders in both themes at AA contrast
Phase 1: Public website core	3-8	All public routes, hero and content bands, programme microsites, news and gallery, Urdu key pages, metadata and JSON-LD, sitemap and Search Console, GBP claimed	M2: public beta deployed on production domain; Lighthouse mobile 90+ across five key pages; office staff can publish a notice unaided
Phase 2: Admissions and results engines	7-12	Five-step application with autosave and uploads, admin review queue and status machine, merit list generation and publication, WhatsApp receipts and broadcasts, results import pipeline and public lookup	M3: end-to-end rehearsal with 100 dummy applications and a real previous-year result file executed before the admission season
Phase 3: Portals, LMS and alumni	11-15	Student and teacher portals (attendance, assignments, gradebook, timetable, resources), admin dashboard analytics, alumni registration, verification and directory, mentorship matching	Two weeks of daily use by one pilot class and three teachers, with attendance-marking under one minute verified
Phase 4: PWA, SEO hardening and launch	14-16	Service worker and offline shell, manifest and install flows, web push, performance and accessibility sweep against budgets, security review, content load for launch, staff training and handover pack	M4: public launch; every budget and gate in Chapters 10 and 11 green on production

Table 17: Phase deliverables and the milestone gate that ends each phase.

12.3 Weekly Cadence

The working rhythm is a fixed weekly loop: Monday planning against the phase board, Tuesday through Thursday build with small pull requests merged daily behind CI, Friday reserved for demo polish and a fifteen-minute demonstration to the office, and a Saturday-morning one-hour review with the principal or focal teacher that doubles as acceptance testing. This cadence matters because it converts the school from a client into a collaborator: every week the institution sees progress, and every week the developer receives course-correction while it is still cheap. The continuous QA and content-entry bar in the Gantt reflects this reality: testing and real content are scheduled work, not volunteer work squeezed between features.

12.4 Risk Register

Risk	Likelihood	Impact	Mitigation
Scope creep from enthusiastic stakeholders	High	High	The scope contract of Chapter 3; every addition must displace equal effort or wait for v2; the principal endorses the contract at kickoff
Free-tier ceiling hit (DB, bandwidth, or WhatsApp cost)	Medium	Medium	Chapter 14 monitors limits monthly; upgrade path pre-approved at modest cost; broadcasts batched and templated to stay inside WhatsApp pricing
Institutional data collection slower than code	High	Medium	Content gathering is scheduled inside Phase 1-2 weeks with named owners; placeholder discipline keeps build unblocked; the office demo cadence surfaces blockers weekly
Connectivity and power constraints for the developer	Medium	Medium	Offline-first workflow (local dev against local Postgres); documentation-first habits; GitHub as the single source of truth from day one
Solo-developer single point of failure	Medium	High	Handover pack and admin manual written as phases close, not at the end; runbooks in the repository; a second teacher is trained as power user at launch
Admission season arrives before Phase 2 completes	Medium	High	Phase 2 is deliberately scheduled before the season; fallback is the paper form mirrored digitally with a one-day manual import path

Table 18: Risk register with likelihood, impact, and the mitigation already embedded in this plan.

13. Quality Assurance and Testing

TEST PYRAMID • DEVICE MATRIX • CI GATES

Quality is enforced by machines wherever possible and by scripted human checks wherever not. The test strategy follows a pragmatic pyramid tuned for one developer: a broad base of fast unit tests on pure logic, a focused band of integration tests on data access and policies, a thin layer of end-to-end journeys on Playwright, and a mandatory human pass per milestone against written checklists.

13.1 Test Pyramid

- Unit (Vitest): merit-list ranking, eligibility rules, attendance aggregation, notification trigger logic, and every date/time boundary, targeting the pure functions that would silently corrupt institutional data if wrong.
- Integration (Vitest + test database): Prisma queries, RLS policy verification under student, teacher, and admin sessions, and server-action validation, including the negative cases that matter most (a student requesting another student's marks must receive nothing).
- End-to-end (Playwright): the five golden journeys of Chapter 4, executed against a seeded staging project on every release candidate: apply for admission, look up a result, mark attendance, submit and grade an assignment, and register as alumni.
- Accessibility (axe in Playwright): zero serious or critical violations on the eleven template routes, both themes.
- Visual regression (Playwright screenshots): the style-guide route and home page in both themes, guarding the design system against accidental drift.

13.2 Device and Browser Matrix

Tier	Devices / browsers	Test depth
Primary (every PR)	Chrome Android (emulated mid-range), desktop Chrome, Safari iOS (physical device borrowed for milestone weeks)	Full golden journeys
Secondary (milestones)	Samsung Internet, Firefox Android, desktop Edge and Firefox, Safari macOS	Key pages and one journey
Constraint cases (milestones)	Slow 3G throttling, 2 GB RAM emulation, offline and flaky-network profiles, 200 percent zoom, forced RTL	Performance, offline, and accessibility spot checks

Table 19: Device and browser matrix with the depth each tier receives.

13.3 Continuous Integration Gates

Every pull request runs the full gate sequence in GitHub Actions: lint, type-check, unit and integration suites, build, Lighthouse CI against a static preview with budgets from Chapter 10 asserted as pass/fail, and the axe accessibility scan. Merges to main deploy a Vercel preview for the weekly school demonstration, and the production deployment happens only from tagged releases, so launch day is a promotion of an already-tested artifact rather than a fresh adventure. Each milestone additionally runs the human checklist: the office staff publish real content unaided, a teacher marks real attendance, a parent follows the WhatsApp message end to end, and the principal signs the gate.

14. Deployment, DevOps and Cost

PIPELINE · ENVIRONMENTS · FREE-TIER ECONOMICS · MONITORING

The delivery philosophy is: one repository, three environments, zero servers, and a cost model that starts at exactly zero rupees per month. This chapter fixes the pipeline, the environment discipline, the exact free-tier ceilings and their monitoring, and the pre-approved upgrade path so that growth never becomes an emergency.

14.1 Pipeline and Environments

GitHub holds main plus feature branches with branch protection on main. Every push to a feature branch produces a Vercel preview deployment with its own URL and its own seeded Supabase branch database for integration tests; every merge to main after the CI gates redeploys the staging project; and production deploys only from version tags (v1.0.0 at launch) with Vercel's instant rollback as the safety net. Environment variables are segregated into preview, staging, and production scopes in both Vercel and Supabase, with production secrets write-restricted to the maintainer account, and the whole promotion path is documented as a runbook in the handover pack so that a successor developer can operate it on day one.

Variable	Scope	Notes
DATABASE_URL / DIRECT_URL	All scopes	Prisma pooled plus direct connections; production points at the Supabase pooler
NEXT_PUBLIC_SUPABASE_URL / ANON_KEY	All scopes	Public by design; all real authorisation happens in RLS
SUPABASE_SERVICE_ROLE_KEY	Server only	Used exclusively by the import pipeline and cron jobs; never shipped to the client
CLOUDINARY_* (cloud, key, secret)	Server + upload widget signature route	Signed uploads from admin; delivery uses public URL patterns
RESEND_API_KEY	Server only	Transactional email
WHATSAPP_TOKEN / PHONE_ID	Server only	Cloud API; templates pre-approved with Meta
CRON_SECRET	Server only	Authenticates scheduled function invocations

Table 20: Environment variable contract across the three deployment scopes.

14.2 Free-Tier Economics

Service	Free-tier ceiling (as researched)	Launch-scale estimate	Headroom plan
Vercel Hobby	100 GB bandwidth/month; unlimited static requests; generous serverless allowances for non-commercial use	8-15 GB/month at 6k sessions	Pro tier at ~\$20/month only if bandwidth doubles; non-commercial school use stays compliant on Hobby
Supabase Free	500 MB database, 1 GB file storage, 50k MAU, 2 projects, 5 GB egress; projects pause after 1 week of inactivity	~120 MB DB, ~600 MB storage at year one	Monitor monthly; upgrade to Pro ~\$25/month for 8 GB DB and no pausing when students exceed ~800 active
Cloudinary Free	~25 credits/month (about 25 GB bandwidth and 25k transformations)	12-18 GB transformed delivery	Paid tier or image budget diet (larger LQIP, fewer gallery resolutions) at 2x growth
GitHub Free	2,000 CI minutes/month	300-500 minutes/month	Self-hosted runner on the school's own PC if ever needed
Resend Free	3,000 emails/month (100/day)	~800/month	Domain-verified digest batching keeps well inside
WhatsApp Cloud API	Per-message utility pricing, tiny at alert volumes	~PKR 2-6k/year at 300 opted-in families	Budgeted from school operating costs; templated and batched

Table 21: Free-tier ceilings, realistic launch-scale estimates, and the pre-approved headroom path per service.

The honest total cost of ownership for year one is therefore the WhatsApp messaging spend, the domain renewal, and the developer's time; every other line is zero rupees at the usage levels this plan projects. The upgrade triggers are written as monitoring thresholds rather than emergencies: the maintainer checks the limits dashboard on the first Saturday of each month, and any service crossing 70 percent of its ceiling triggers the documented upgrade decision while the family of options is still calm.

14.3 Monitoring and Operations

- Vercel Analytics for traffic and Web Vitals in the field, reviewed in the Saturday session alongside Search Console queries and Clarity heatmaps.
- Supabase dashboard for database size, auth activity, and storage growth against the monthly thresholds.
- UptimeRobot free checks on the home page, the results lookup, and the portal login, with WhatsApp alerts to the maintainer and the office focal teacher.
- Error tracking through Vercel log drains into a small daily digest, so failures surface as routine maintenance rather than surprises.
- A monthly operations runbook checklist in the repository: backup restore rehearsal, dependency updates with CI verification, expired-content sweep, and access review of staff accounts.

15. Maintenance and Future Evolution

CONTENT OPS · HANDOVER · THE V2 HORIZON

A school platform that is alive in five years will have outlived most of its contemporaries, and that outcome is decided by boring habits, not by brilliant features. This chapter defines the operating rhythm after launch, the handover that removes dependence on any single person, and the version-two horizon that gives stakeholders a place to send their good ideas instead of into the current sprint.

15.1 Content Operations

The editorial calendar from Chapter 10 continues as a weekly habit owned by a named teacher-editor with admin rights: one story or notice batch per week in term time, plus the four seasonal peaks (admission opening, result days, prize distribution, and the culture festival) that each get a small planned package of stories, photographs, and a WhatsApp broadcast. Structural maintenance follows the school's clock: term-start timetable and roster updates, annual faculty and fee-structure refreshes before admission season, and a yearly results-archive snapshot that is also the trigger for the backup-restore rehearsal. The Saturday one-hour review continues at a lighter monthly cadence after the first term, keeping the institution in the loop while returning the developer's weekends.

15.2 Handover and Institutional Memory

- The handover pack ships at M4 and lives in the repository: architecture decision records explaining why each choice was made, the operations runbook from Chapter 14, the admin manual with step-by-step screenshots in English and Urdu, and a short recorded walkthrough in Urdu of the five tasks office staff perform most.
- One teacher is trained as a power user during Phase 4 and co-runs the first month of content operations, so at least two people can publish, review admissions, and trigger broadcasts.
- All credentials use institutional email ownership with recovery codes lodged with the principal's office in a sealed envelope, ending personal-account lock-in.
- The style-guide route and the design tokens remain the single source of visual truth, so future contributors extend the system rather than fork it.

15.3 The Version-Two Horizon

The deferred list in Chapter 3 becomes the v2 backlog, sequenced by community value: full Urdu and Pashto localisation of every route; online fee payment with e-receipts, following the TMUC precedent; a deeper LMS with video lessons and auto-graded quizzes to support the ESEF-era push toward digital classrooms in KPK; an AI study assistant grounded in the school's own resources; SMS as a fallback notification channel for families without WhatsApp; and, if district demand appears, a public open-data view of anonymised result statistics that would make GHSS Ghallanai a transparency reference for the province. Each of these is a deliberate post-launch decision to be made with a year of real usage data in hand, by which time the platform's telemetry will argue better than any plan could.

The measure of this plan is not the launch-day screenshot. It is the morning, two years from now, when a mother in Ekka Ghund opens a home-screen icon, reads her daughter's result in Urdu, and the school has become a little more present in that family's life than it was before.

Appendix A: Research Sources

THE SEVEN INSTITUTIONS EXAMINED PAGE BY PAGE

The competitive analysis in Chapter 2 was assembled from direct page-level examination of the following institutions during the preparation of this plan, using their live websites as primary sources. Extracted patterns are summarised with each entry; the design decisions of this plan cite them throughout.

Institution	URL	What was extracted
Aitchison College, Lahore (1886)	aitchison.edu.pk	Video hero pattern; heritage IA (History, Archives, Principal's Office, Code of Conduct); Aitchison Connect alumni portal; Teaching and Learning Centre; university counselling and placements; IBM Plex Sans + Montserrat + Playball typography; Bootstrap/Swiper stack analysis
TMUC, Islamabad	tmuc.edu.pk	Mega-menu programme navigation; AOS scroll reveals, odometer counters, slick carousel; Hind + Lexend Deca typography; online fee payment; career development centre and incubation; WhatsApp click-to-chat; The Events Calendar usage
Phillips Exeter Academy, USA (1781)	exeter.edu	2025 CASE Circle of Excellence award-winning redesign; narrative homepage; typographic statistics (30M aid, 12:1 ratio); Harkness branded storytelling page; immersive YouVisit tour; #iamExeter story format; need-blind admissions messaging
Eton College, UK (1440)	etoncollege.com	Eight-section mega-menu over 60+ pages; Parent Portal separation; EtonX online learning; OEA alumni network; museums and facilities-for-hire; PWA manifest and apple-touch icons; MS Clarity + GTM analytics; podcast and term-dates surfaces
Punjab Colleges / PGC, Lahore (1985)	pgc.edu	Network-scale programme taxonomy matching GHSS (FA, F.Sc both tracks, ICS, ICom); achievements counters (142 top positions 2024); alumni registration journey; student portal in header; scholarship and fee-concession presentation for parents
Islamia College Peshawar (1913)	icp.edu.pk	Government-side reference: CMS and eOffice integrations; online FA/FSc application forms; downloadable prospectus and fee documents; alumni association; QEC and ORIC presence; press-release channel; legacy front-end gap analysis
Edwardes College Peshawar (1900)	edwardes.edu.pk	Online admissions e-Portal and online merit list; alumni registration; Urdu Nastaliq typography; Pashto Culture Day and video tour content; counseling and leave-application workflows; notice board and societies model

Table 22: Research source register with the specific patterns extracted from each institution.

Appendix B: Master Build Checklist

PRINT ME • PIN ME • TICK ME

This checklist compresses the whole plan into the order work actually happens. Each item is phrased as a verifiable action, and each maps back to the chapter that specifies it in full, so any item that raises questions has a one-hop route to its answer.

B.1 Phase 0 - Foundation (weeks 1-3)

1. Create the GitHub repository with branch protection and the folder skeleton of Chapter 8.4.
2. Provision the Supabase project, run Prisma migrations for schema v1, and verify RLS is enabled by default.
3. Implement design tokens (Chapter 5.2) and the theme architecture (Chapter 5.3) with the no-flash inline script.
4. Build the component library and the style-guide route, verifying AA contrast in both themes.
5. Produce the crest logo, icon set with duotone gold treatment, and PWA icon exports.
6. Wireframe every template from Chapter 6 and review them with the principal's office at week 3 (M1 gate).

B.2 Phase 1 - Public Core (weeks 3-8)

1. Build all public routes per the registry (Chapter 4.2), starting with home, programmes, and admissions guidance.
2. Implement the hero (poster-first video), ticker, odometer stats band, and all content bands.
3. Author metadata, JSON-LD (five vocabularies), dynamic sitemap, and robots; verify in Search Console and claim the Google Business Profile.
4. Load real content: history, principal's message, faculty, fee structure, FAQs, facilities; gather photographs with consent flags.
5. Publish Urdu renderings of home, admissions, fees, results, and contact with Nastaliq typography.
6. Run Lighthouse CI budgets on five key pages and pass 90+ mobile; demo notice publishing to office staff (M2 gate).

B.3 Phase 2 - Admissions and Results (weeks 7-12)

1. Build the five-step application with autosave, uploads, and zod validation; issue application numbers and receipts.

2. Implement the admin review queue, status machine, and audit logging.
3. Implement merit-list generation, HTML/PDF publication, and the WhatsApp broadcast integration with approved templates.
4. Stand up the supervised results import pipeline with the two-person checksum confirmation.
5. Ship the public result lookup with rate limiting and the toppers wall.
6. Rehearse end-to-end with 100 dummy applications and one real previous-year result file before the season opens (M3 gate).

B.4 Phase 3 - Portals and Alumni (weeks 11-15)

1. Build the student and teacher portals with attendance, assignments, gradebook, timetable, and resources.
2. Verify RLS under all three roles with negative tests (student cannot read another student's rows).
3. Build the admin dashboard: content management, admissions queue, user management, and analytics views.
4. Ship alumni registration, human verification workflow, directory, stories, and mentorship matching.
5. Run the two-week pilot with one class and three teachers; confirm the one-minute attendance mark.

B.5 Phase 4 - PWA, Hardening and Launch (weeks 14-16)

1. Generate the service worker with the three caching strategies and the offline page; verify offline shell and update toast.
2. Finalise the manifest, maskable icons, shortcuts, and install prompts; test on Android and iOS Safari.
3. Enable web push with topic flags and quiet hours; rehearse a results-day broadcast.
4. Run the full performance, accessibility (axe zero-serious), and security sweep (CSP headers, rate limits, secrets audit).
5. Complete content load, the handover pack, the admin manual in English and Urdu, and staff training.
6. Tag v1.0.0, promote to production, and monitor the first result-day traffic spike from the operations runbook (M4 gate: launch).